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Synergistic promotion of straw anaerobic digestion by a composite microbial consortium and micro-aeration: Enhancing lignocellulose decomposition and methanogenesis

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ABSTRACT

The stable structure of lignocellulose is the main obstacle to the anaerobic digestion of agricultural straw. In this study, a composite microbial consortium composed of *Phanerochaete chrysosporium* and *Aspergillus niger*, coupled with intermittent micro-aeration, was employed to enhance lignocellulose decomposition and methanogenesis. Under an optimal micro-aeration intensity of 4 % (V_{air}:$V_{reactor}$/day), methane production was 22.57 % higher than that under strict anaerobic conditions. A higher aeration intensity (6 %) further improved organic matter removal, but decreased methane production. Micro-aeration induced periodic fluctuations in oxidation-reduction potential during anaerobic digestion to form reactive oxygen species, which significantly enhanced the activities of lignin peroxidase (2.92-fold), manganese peroxidase (1.99-fold), and laccase (2.83-fold), thereby accelerating lignin decomposition and humus formation. The resulting humic substances acted as natural electron shuttles, facilitating microbial electron transfer and elevating intracellular NADH/NAD^+ ratios, which ultimately stimulated methanogenesis. Microbial community analysis confirmed that the inoculated fungi survived and remained metabolically active under micro-aerobic conditions, effectively degrading lignocellulose to provide accessible substrates for methanogens. This study demonstrated that integrating a composite fungal consortium with controlled micro-aeration could be a robust and efficient strategy for valorizing straw waste.

1. Introduction

Anaerobic digestion is an efficient approach for agricultural waste management by converting organic solids into methane or valuable chemicals such as fatty acids. However, agricultural residues like crop straw mainly consist of lignocellulose—a complex and chemically stable mixture of lignin, cellulose, and hemicellulose—which resists microbial breakdown and thus limits the digestion efficiency [1,2]. *Phanerochaete chrysosporium*, an aerobic lignin-decomposing fungus, secretes extracellular enzymes such as lignin peroxidase (LiP), manganese peroxidase (MnP), and laccase (Lac) to decompose lignin [3]. Specifically, oxidases and dismutases in *P. chrysosporium* catalyze oxygen reductions: first, O_2 accepts two electrons to form superoxide anions ($\bullet O_2^-$; $O_2 + 2e^- \rightarrow \bullet O_2^-$), which then disproportionate in the presence of protons to generate hydrogen peroxide ($2\bullet O_2^- + 2H^+ \rightarrow H_2O_2 + O_2$). This initiates Fenton-like reactions ($Fe^{2+} + H_2O_2 \rightarrow Fe^{3+} + \bullet OH + OH^-$) [4], where hydroxyl radicals ($\bullet$OH) cleave aromatic structures and β-O-4 ether bonds in lignin, disrupting its protective matrix and improving the accessibility of cellulose for enzymatic hydrolysis [5–7]. Additionally, phenolic compounds derived from lignin decomposition undergo enzyme-catalyzed polymerization and humification [8,9]. Owing to these properties, *Phanerochaete chrysosporium* has been widely applied in aerobic composting and anaerobic pretreatment processes [10,11].

Methanogenesis requires strict anaerobic conditions. However, recent research has demonstrated that controlled micro-aeration in anaerobic digestion systems can promote the enrichment of hydrolytic and acidogenic microorganisms (including actinomycetes and fungi) without inhibiting methanogens [12–14]. Notably, Gong et al. recently reported that *Phanerochaete chrysosporium* secreted ligninolytic enzymes under intermittent micro-aeration, thereby accelerating straw decomposition and enhancing digestion efficiency [15].

While *Phanerochaete chrysosporium* primarily targets lignin decomposition, its effectiveness on cellulose and hemicellulose is limited. *Aspergillus niger*, however, is capable of breaking cellulose into

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cellobiose, which may be converted into glucose via β-glucosidase. Additionally, *Aspergillus niger* secretes various hydrolytic enzymes that decompose hemicellulose's complex backbone and side chains, releasing monosaccharides like xylo-oligosaccharides [16–18]. Therefore, a composite microbial consortium composed of *Phanerochaete chrysosporium* and *Aspergillus niger* is hypothesized to achieve synergistic decomposition of straw lignocellulose. Notably, composite fungi have been widely used in aerobic composting, but there are few studies on their direct application in anaerobic digestion [19].Additionally, controlled micro-aeration is known to improve lignocellulose decomposition by facilitating the activity of oxidative enzymes and promoting the generation of reactive oxygen species (ROS) [20]. Intermittent micro-aeration has been previously reported to minimize the inhibitory effects of excessive oxygen on strictly anaerobic methanogens while enhancing lignocellulose decomposition [12,21]. Therefore, it is assumed that integrating a composite microbial consortium with optimized intermittent micro-aerobic conditions might further enhance straw decomposition and methane production. So far, no reports have been found on the application of composite fungi combined with micro-aeration in anaerobic digestion.

Consequently, this study employed a composite microbial consortium of *Phanerochaete chrysosporium and Aspergillus niger*, combined with intermittent micro-aeration, to enhance lignocellulose decomposition and methanogenesis. The objectives were to evaluate the feasibility of this combined strategy and to elucidate the underlying mechanisms. Specifically, this study investigated how hydroxyl radicals (•OH) disrupt lignocellulosic structures to enhance anaerobic biodegradability. Additionally, the impact of micro-aeration on microbial community was explored. This study was expected to provide a promising strategy to enhance the anaerobic digestion of agricultural waste.

2. Materials and methods

2.1. Experimental materials

The inoculated sludge used in the anaerobic digestion experiments, the sludge from a wastewater treatment plant in Dalian, was stored anaerobically at 4 °C in the laboratory. Its chemical oxygen demand (COD), 48770 ± 413 mg/L. Corn straw was purchased from a farm (Lianyungang, China), chopped, ground into powder, and screened to a particle size of approximately 0.15 mm and used as the sole digestion substrate, without any additional carbon sources. The basic characteristics of the corn straw and the sludge are presented in Table S1. The mixed sludge was activated for 5 days prior to straw addition. *Phanerochaete chrysosporium* and *Aspergillus niger* were obtained from Beina Bio, Henan Industrial Microbial Strain Engineering Research Center. The fungi were activated on potato dextrose agar (PDA) and incubated at 30 °C for 5 days. Colonies were scraped and rinsed with sterile water to obtain spore suspensions. The suspensions were diluted to 2×10^8 spores/mL based on plate count results (colony-forming units, CFU) and stored at −80 °C for subsequent use.

2.2. Experimental design

The anaerobic digestion of corn straw was conducted in batch modes. Each reactor had an effective working volume of 200 mL (inner diameter: 60 mm; height: 70 mm), which was equipped with two sampling ports at the top for feeding or discharge, and biogas collection. A 500 mL gas sampling bag was connected for biogas collection. Batch tests were first conducted to evaluate the effects of microbial consortium and micro-aeration on biogas production. Each reactor was filled with 100 mL of mixed sludge (see Section 2.1), 20 mL of spore suspension (10 mL per strain), and 80 mL of distilled water. Each reactor received 6 g of dry corn straw. The ratio of sludge to spore suspension (100 mL:20 mL) was determined based on preliminary batch tests (data not shown) aimed at optimizing fungal activity without causing system acidification or instability.

The experiments were divided into three groups: the control group (no microbial inoculum, anaerobic conditions), the anaerobic inoculation group, and the micro-aerobic inoculation group with air volumes of 5 mL (air volume ratio [$V_{air}/V_{reactor}$] = 2 %). Subsequently, the effects of aeration intensity on methane production were evaluated under comparable conditions: each reactor was loaded with 100 mL of mixed sludge, 20 mL of spore suspension (10 mL per strain), and 80 mL of distilled water. Every 3 days, 10 mL of substrate (containing 2 g of straw) was fed and an equal volume of digestate was discharged to maintain a hydraulic retention time (HRT) of 60 days. Four aeration groups were tested, with air volumes of 0 (control), 5, 10, and 15 mL (air volume ratio [$V_{air}/V_{reactor}$] = 0 %, 2 %, 4 %, 6 %), designated as Control, 2 % Air, 4 % Air, and 6 % Air. Air was injected into the reactors via a 20-mL syringe through the feeding port, with a single injection every 24 h. The injection process lasted for approximately 5 s to ensure diffusion. All reactors were operated at 37 °C for 60 days.

To address questions regarding the causal link of the fungi and the mechanism of inhibition at aeration, supplementary batch experiments for mechanistic validation was conducted (see Supporting Material for details).

2.3. Analytical methods

Chemical oxygen demand (COD) was measured by standard methods. Methane production was determined by gas chromatography with a thermal conductivity detector (Tianmei GC-7900, China), equipped with a TCD and a GDX — 102/80–100 mesh packed column. Nitrogen was used as the carrier gas at 40 mL/min. The temperatures of the injector, column, and detector were set at 140 °C, 220 °C, and 160 °C, respectively. The morphology of sludge and straw was examined using scanning electron microscopy (SEM, HITACHI SU5000, China) [22]. Sludge electrochemical properties (cyclic voltammetry [CV] and electrochemical impedance spectroscopy [EIS]) were analyzed (see Supporting Material). Electron accepting capacity (EAC) and electron donating capacity (EDC) were used to evaluate the electron accepting/donating capacity of extracellular polymeric substances (EPS) [23].

The activities of lignin peroxidase (LiP), manganese peroxidase (MnP), and laccase (Lac) were determined using enzyme-linked immunosorbent assay (ELISA) kits. Cellulase activity was measured by previous methods (see Supporting Material). Lignin, cellulose, and hemicellulose contents were determined following the Van Soest method [24].

ROS were detected using EPR spectroscopy (EMXPlus, Bruker A300, Germany). 2,2,6,6-Tetramethylpiperidine (TEMP) and 5,5-Dimethyl-1-pyrroline-N-oxide (DMPO) were used as spin traps for $•O_2^-$ and •OH, respectively. Samples were ultrasonically dispersed in water or methanol, mixed with spin traps, and analyzed using capillaries in the EPR instrument. Superoxide anion radical ($•O_2^-$) was quantified using the 2,3-bis(2-methoxy-4-nitro-5-sulfophenyl)-5-carboxypyridine tetrazolium [XTT] assay. Hydroxyl radicals (•OH) were quantitatively analyzed using high-performance liquid chromatography (HPLC) by trapping them with dimethyl sulfoxide (DMSO), which reacts to generate formaldehyde (HCHO), following the procedure outlined in Text S1.

Oxidation-reduction potential (ORP) and headspace oxygen were measured using an ORP electrode (Model STARTER 3100, USA) and a fiber optic oxygen transmitter (FiBox3, PreSens GmbH, Germany). These parameters, rather than dissolved oxygen (DO), were selected as they more accurately reflect the bulk redox conditions and the transient oxygen exposure in the anaerobic headspace. Each group of 250 mL reactors was tested continuously for 24 h, Excitation-emission matrix (EEM) fluorescence spectra were obtained using an Agilent Cary Eclipse spectrophotometer (Santa Clara, CA, USA). Fourier-transform infrared spectroscopy (FTIR) analysis was conducted using a Thermo Nicolet IS50 spectrometer (scan range: 4000–400 cm^{-1}, resolution: 4 cm^{-1}).

2.4. Microbial community analysis

At the end of the experiment, sludge samples were collected for high-throughput 16S rRNA sequencing. Microbial DNA was extracted using a rapid soil DNA extraction kit. The V3–V4 regions of archaea and bacteria, and the ITS region of fungi were amplified by PCR. Purified PCR products were used to construct libraries with the Illumina TruSeq DNA protocol and sequenced on an Illumina TruSeq 2000 platform. The sequencing data have been deposited in the NCBI Sequence Read Archive (SRA) under BioProject accession number PRJNA1367766.

2.5. Statistical analysis

All experiments were conducted in triplicate, and data are presented as mean ± standard deviation unless otherwise noted. Statistical comparisons between groups were performed using one-way analysis of variance (ANOVA) followed by SPSS 22.0. A p-value < 0.05 was considered statistically significant.

3. Results and discussion

3.1. Enhancement of anaerobic digestion performance

3.1.1. Effects of the microbial consortium and micro-aeration

In sequential batch anaerobic digesters using straw as the substrate, a microbial consortium of *Phanerochaete chrysosporium* and *Aspergillus niger* was integrated with micro-aeration to evaluate the performance of the digesters. The results clearly indicated that the benefits of the fungal consortium were oxygen-dependent. Under strictly anaerobic conditions without aeration (Fig. S1), the consortium provided negligible benefits; methane production in the anaerobic inoculation group (218.9 mL) was marginally higher than in the anaerobic control group (205.7 mL). In sharp contrast, under micro-aerobic conditions (2 % air), the consortium increased methane production to 247.3 mL, a 20.2 % increase over the anaerobic control group. This demonstrated that micro-aeration is essential to activate the metabolic functions of the inoculated fungi for enhancing anaerobic digestion.

3.1.2. Effect of aeration intensity

Under semi-continuous feeding with a single-dose fungal consortium inoculation, the effects of aeration intensity on methane production were investigated. Results (Fig. 1A, B) revealed that the 4 % air group achieved the highest methane production, which was 15.93 % higher than the no-air control group (17.75 vs. 15.31 mL/g VS·day). compared to the no-air control group, the 2 % and 6 % aeration groups increased by 13.78 % (17.42 ± 0.96 mL/g VS·day) and 13.12 % (17.32 ± 0.98 mL/g VS·day), respectively. Cumulative methane production reached 1392 mL (2 % air group), 1607 mL (4 % air group), and 1465 mL (6 % air group), which were 6.17 %, 22.57 %, and 11.75 % higher than the control (1311 mL), respectively. This 22.57 % enhancement must be benchmarked against other recent strategies for straw valorization. For instance, in-situ additives, such as nano-bubble technology, have reported comparable gains (24.08 %) [25]. In contrast, dedicated, energy-intensive methods like hydrothermal pretreatment could achieve significantly higher yields, reaching 152.28 % [26]. However, the primary advantage of the in-situ approach lied not in achieving the absolute maximum yield, but in avoiding the high capital and energy costs associated with a separate pretreatment step. The value of this method was the operational simplicity. The total chemical oxygen demand (tCOD) decreased in the aerated groups (Fig. 1C, D), which were 1.39 × 10^6 mg/L at 2 % aeration, 1.35 × 10^6 mg/L at 4 % aeration, and 1.30 × 10^6 mg/L at 6 % aeration. These represented reductions of 4.36 %, 7.20 %, and 8.91 % compared with the control group (1.45 × 10^6 mg/L), respectively. Conversely, the soluble chemical oxygen demand (sCOD) levels in the aerated groups were 2341 mg/L at 2 % aeration, 2533 mg/L at 4 % aeration, and 2331 mg/L at 6 % aeration, all of which were higher than that of the control group (2255 mg/L), indicating that micro-aeration enhanced the hydrolysis and acidogenesis of lignocellulose to increase the production of soluble carbon compounds [27,28]. Collectively, the results confirmed that the combination of the composite

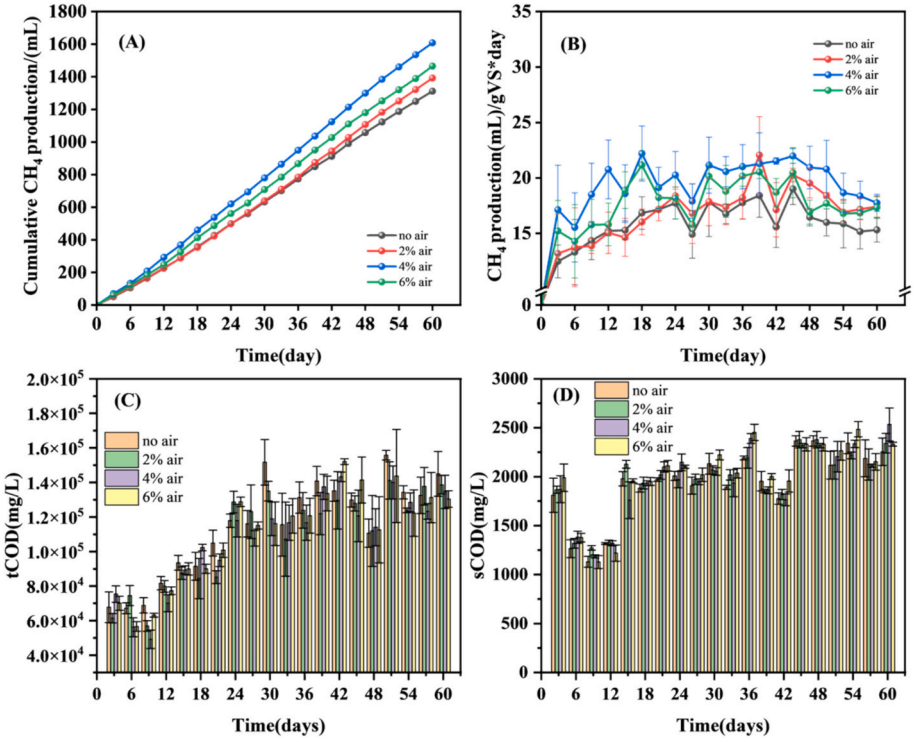


Fig. 1. The performances of the methane production in biogas and the chemical oxygen demand (COD) in the effluent. (A) Daily average methane production, (B) cumulative methane production, (C) tCOD in the effluent, (D) sCOD in the effluent.

microbial consortium and micro-aeration promoted the decomposition of organic matter and enhanced methane production. Although the cumulative methane production had not reached a complete plateau by day 60 (Fig. 1A), the experiment was terminated as the daily methane production rates (Fig. 1B) and COD removal (Fig. 1C, D) in the semi-continuous reactors had reached a relatively stable state, allowing for a robust comparison of performance under the different aeration intensities.

It is noteworthy that while the 6 % air group in the semi-continuous run exhibited high tCOD removal, its methane production was inhibited. To validate the mechanism, VFA concentrations were quantified in the supplementary batch experiment. The results (presented in Fig. S4) confirmed this hypothesis: The 4 % aeration group (Fungi-MA) maintained a low VFA concentration [265.57–767.50 mg/L]. In contrast, the 6 % aeration group (Fungi-MA-6 %) showed the VFA accumulation, reaching [313.59–888.94 mg/L], primarily composed of acetate and propionate.

This evidence strongly supported that excessive aeration (6 %) caused a "decoupling" of hydrolysis from methanogenesis, where the oxygen inhibited VFA-consuming methanogens, leading to VFA accumulation and the observed drop in methane production.

3.2. Lignocellulose decomposition

Micro-aeration significantly enhanced lignocellulose removal (Fig. 2A), as evidenced by decreasing digestate residue levels in the following order: no air > 6 % air > 2 % air > 4 % air. The 4 % air group exhibited the lowest lignocellulose content (52.06 %: hemicellulose with 15.34 %, cellulose with 29.90 %, lignin with 7.71 %). This was followed by the 2 % air group (53.63 %: 14.87 %, 31.11 %, 7.69 %, respectively) and the 6 % air group (55.05 %: 16.04 %, 30.16 %, 8.84 %, respectively). The control group retained the highest lignocellulose content (59.99 %: 21.38 %, 29.92 %, 8.68 %). The lignocellulose residue in the 4 % air group was 7.93 percentage point lower than that in the control group. These results confirmed that the combination of the composite microbial consortium and micro-aeration enhanced lignocellulose decomposition.

Micro-aeration also increased the activity of lignocellulolytic enzymes. Laccase activity was highest in the 4 % air group (0.51 U/mL), 2.83-fold higher than the control group (0.18 U/mL). Other enzymes peaked in the 6 % air group: lignin peroxidase (LiP, 0.18 U/mL vs. 0.06 U/mL), manganese peroxidase (MnP, 2.76 U/mL vs. 1.39 U/mL), and cellobiohydrolase (0.19 U/mL vs. 0.16 U/mL). These activation levels are significant, as previously noted that [29].Overall, the activities of lignocellulose-degrading enzymes were significantly higher in the micro-aerated groups than in the control group (Fig. 2B–E), indicating that micro-aerobic conditions promoted the secretion of key oxidative enzymes involved in lignocellulose breakdown [30,31]. Moreover, the enzymatic activity exhibited a positive correlation with oxygen concentration, thereby accelerating the decomposition of lignocellulose and facilitating the release of sCOD for the subsequent methanogenesis [2].

Accordingly, morphological analysis of straw digestate (Fig. S2) showed that the residues of the control group was of compact structure, while the residues of the three aerated groups were clearly fragmented. Besides, the aerated groups exhibited dense fungal mycelia and macroscopic fungal aggregates, which was the result of the active fungal metabolism. The aggregates observed in the aerated groups might have improved microbial tolerance to oxygen stress [32].

3.3. Redox potential and ROS

Since these lignocellulolytic enzymes require oxidative triggers, it was hypothesized that micro-aeration induced Reactive Oxygen Species (ROS). Oxidation-Reduction Potential (ORP) and headspace oxygen concentrations were quantified (Fig. 3A). Oxygen levels increased in the beginning of aeration, and then declined. Higher aeration intensities delayed the oxygen depletion. Specifically, in 2 % air group, the headspace oxygen concentration peaked at 2.52 % at 3.5 h and was depleted at 13.5 h, while in the 6 % air group, the oxygen concentration peaked at 4.25 % at 4.5 h and took 15.5 h to deplete.

Consistently, ORP measurements revealed significant redox

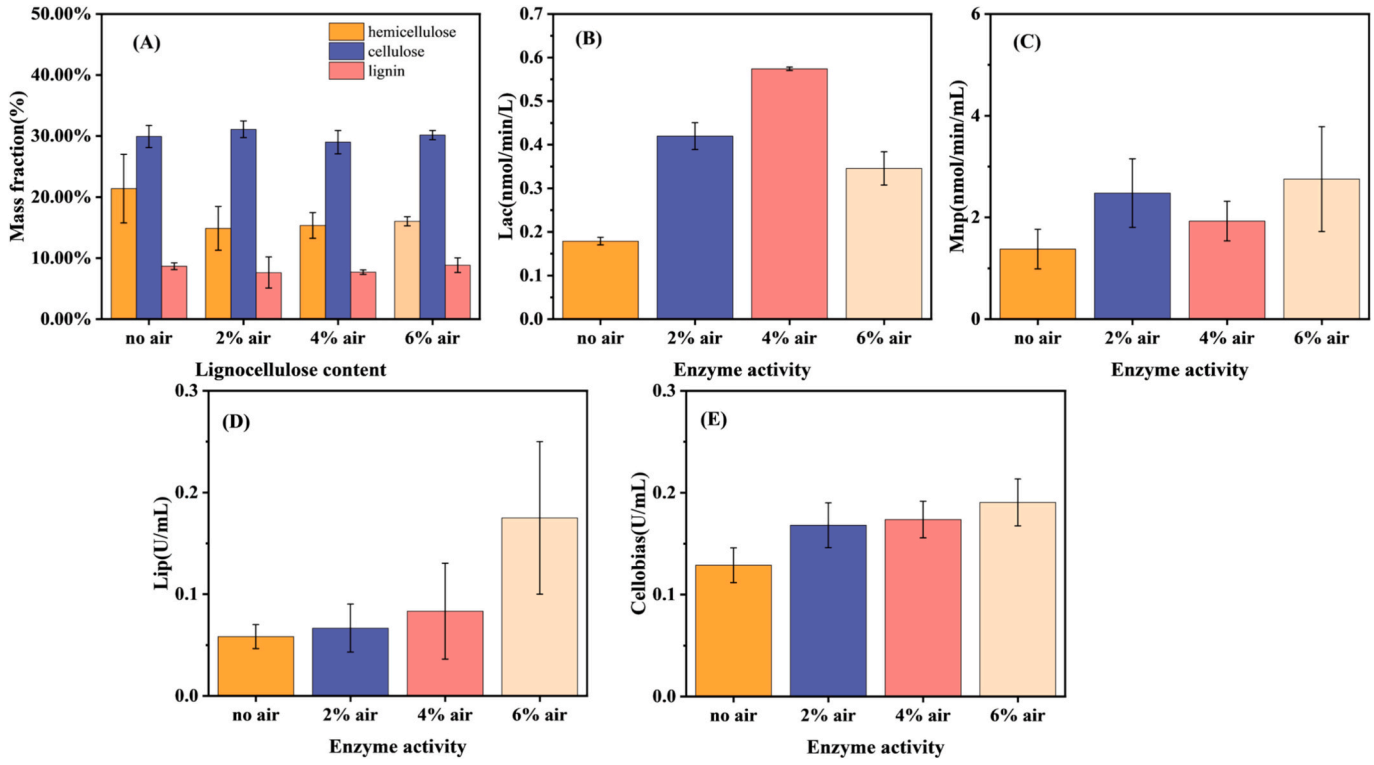


Fig. 2. Lignocellulose decomposition and enzyme activities of straw digestate. (A) The percentage of lignocellulose content; (B) lignin peroxidase activity; (C) manganese peroxidase activity; (D) laccase activity; (E) β-glucosidase enzyme activity.

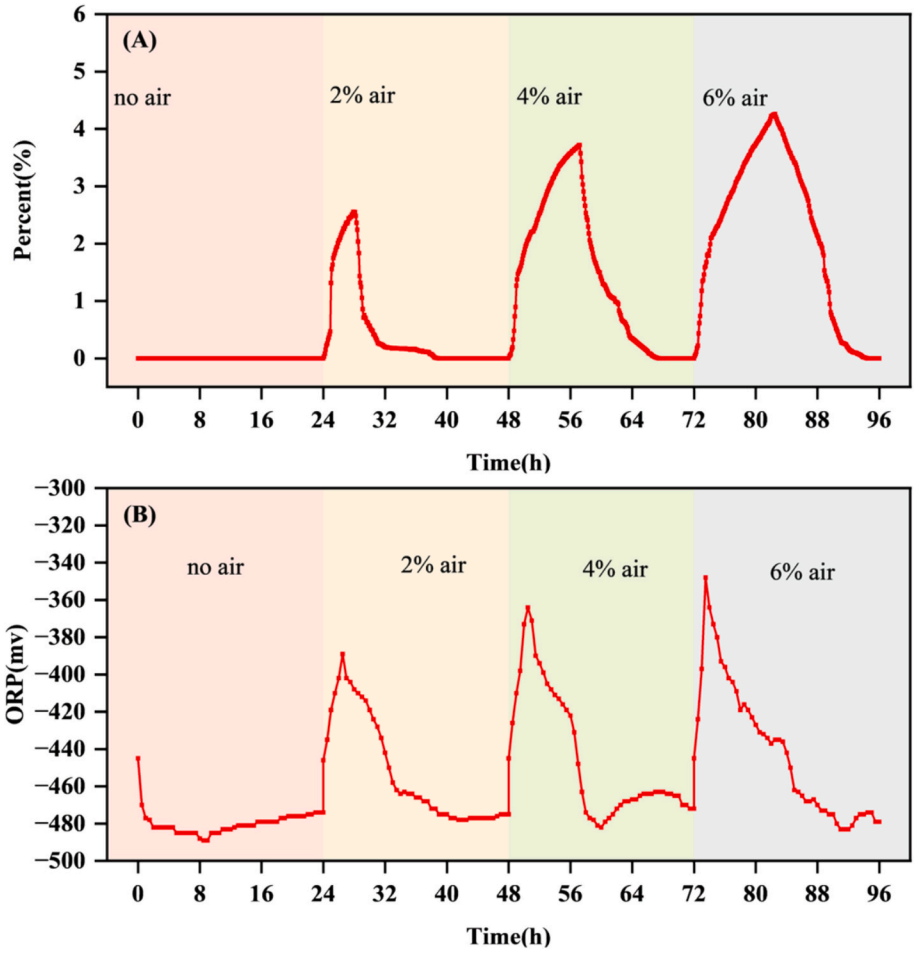


Fig. 3. Redox potential and dissolved oxygen in the reactor (A) time distribution of redox potential; (B) time distribution of dissolved oxygen;

oscillations after aeration. ORP was initially regulated to be the same 450 mV in all groups (almost strictly anaerobic), aeration at 2 %, 4 %, and 6 % increased ORP values to −389, −364, and − 348 mV, respectively. As the headspace oxygen was consumed, ORP gradually returned to the strict anaerobic levels (around −480 to −470 mV). The no-air control group maintained stable ORP (−450 to −480 mV) [33,34], while the aerated groups showed oxygen-dependent fluctuations (Fig. 3B). It has been reported that *Phanerochaete chrysosporium* secrete laccase and peroxidases under aerobic conditions to mediate •OH generation [35]. Specifically, Laccase oxidizes phenolic lignin derivatives to phenoxy radicals, which form quinones [36]. These are reduced to semiquinone radicals by intracellular reductants, releasing superoxide ($•O_2^-$) during quinone regeneration. Subsequent Fenton-like reactions convert $•O_2^-$ to •OH. EPR analysis confirmed that •OH and $•O_2^-$ signals were enhanced in the aerated digestate, which were intensified with the aeration duration (Fig. 4A, B). This confirmed that a composite microbial consortium could trigger Fenton-like reactions under micro-aerobic conditions [37,38] (see Eqs. 1–3).

Quantification of •OH showed that aeration facilitated the •OH generation in the anaerobic system (Fig. 4C). The 4 % air group produced 17.8 % more •OH than the 2 % air group (45.03 vs. 38.23 μmol/L). The 6 % air group showed a 33.3 % increase over the 2 % group (50.96 vs. 38.23 μmol/L). Negligible •OH was detected in the control group. Stronger redox oscillations at the higher aeration condition likely intensified Fenton-like reactions to enhance organic removal but inhibiting methanogenesis, consistent with the lower methane production in 6 % air group.

$$O_2 + e^-_{(microbic)} \rightarrow \dot{O}_2^- \quad (1)$$

$$2\dot{O}_2^- + 2H^+ \rightarrow H_2O_2 + O_2 \quad (2)$$

$$Fe^{2+}_{(Sludge)} + H_2O_2 \rightarrow Fe^{3+} + \dot{O}H + OH^- \quad (3)$$

3.4. Effects of micro-aeration on straw humification

The controlled introduction of air into the anaerobic digestion system could accelerate humification. The ROS-driven enzymatic degradation of lignin generated phenolic intermediates, a process driven by extracellular oxidative enzymes secreted by microorganisms. These enzymes decomposed lignin to produce phenolics and other aromatics, which were then polymerized into complex and stable humic substances [39,40]. Due to its rich aromatic structure, lignin was highly susceptible to enzymatic oxidation, serving as a primary substrate for the generation of humic substances precursors [41–43]. Under micro-aerobic conditions, extracellular enzymes such as laccase (Lac), lignin peroxidase (LiP), and manganese peroxidase (MnP), secreted primarily by *Phanerochaete chrysosporium*, played a central role in initiating these transformations. Specifically, they attacked lignin via distinct catalytic pathways: Lac oxidized phenolic units to phenoxyl radicals; the high-redox-potential LiP directly oxidized non-phenolic rings to cation radicals; and MnP utilized a diffusible Mn(III)-chelate as a redox mediator [44].

The highly reactive radical intermediates generated by these enzymatic actions could directly initiate the depolymerization and structural modification of lignin. It included the preferential cleavage of ether bonds, such as the prevalent β-O-4 linkage, and the oxidation of C-α side chains to form carbonyl and carboxyl groups, thereby releasing phenolic

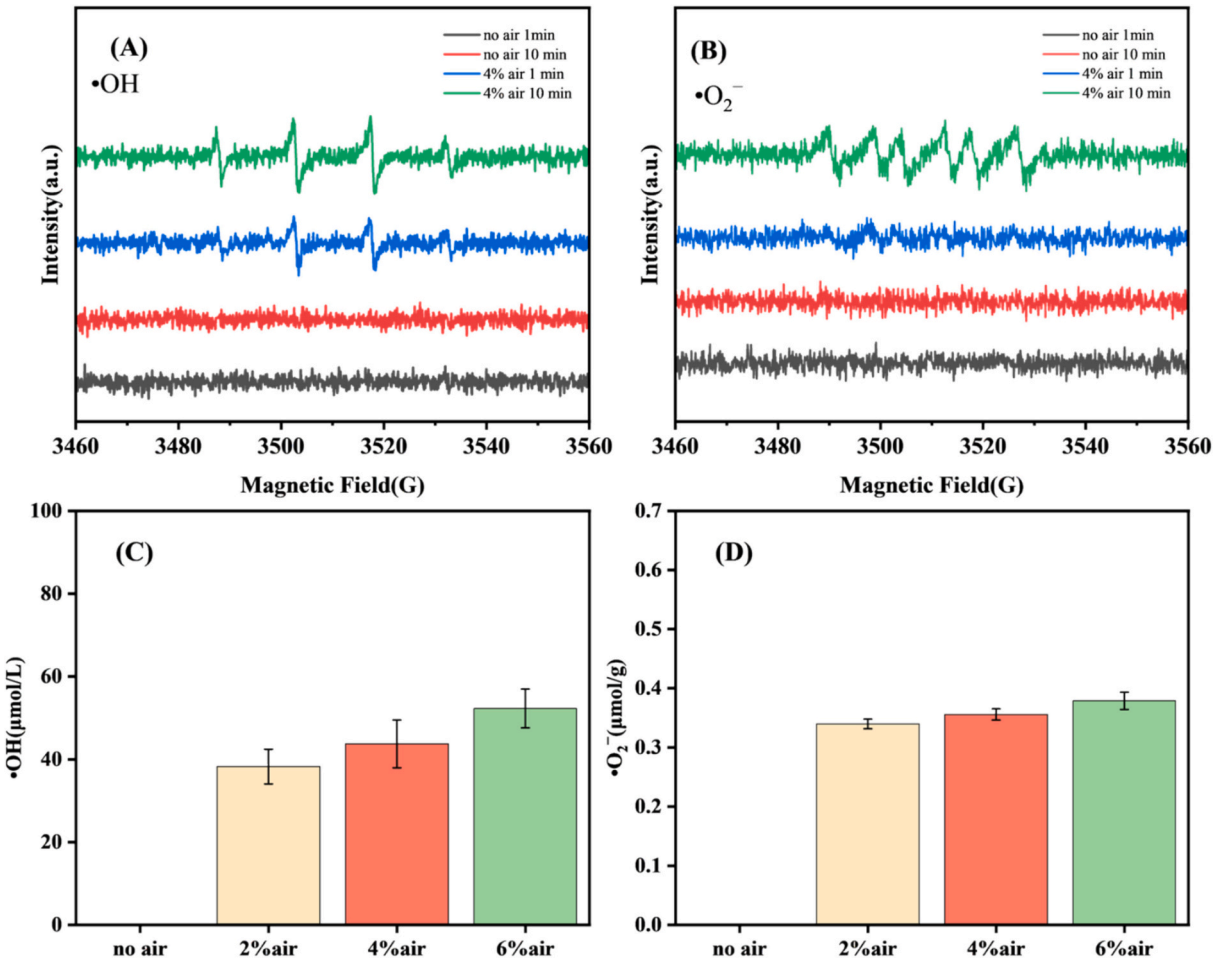


Fig. 4. Detection of reactive oxygen species (A) EPR mapping of •OH; (B) EPR mapping of $•O_2^-$; (C) quantitative analysis of •OH; (D) $•O_2^-$ quantitative analysis.

monomers like vanillin and syringaldehyde. Concurrently, the products and intermediates of these enzymatic reactions (H_2O_2 and quinones/hydroquinones) could engage in subsequent chemical reactions, such as quinone redox cycling and Fenton-like reactions, leading to the indirect generation of highly potent hydroxyl radicals (•OH). As non-selective, powerful oxidants, these hydroxyl radicals further attack and disrupt lignin, promoting aromatic ring-opening to form intermediates such as vanillic acid (Fig. 4A), which served as fundamental building blocks for subsequent humification processes.

FTIR and EEM analysis were conducted to investigate the humification. FTIR spectra (Fig. S3; key absorption bands summarized in Table S2) demonstrated that freeze-dried samples from the aeration groups exhibited distinct structural alterations of lignocellulose after anaerobic digestion. Notably, the intensity of characteristic absorption

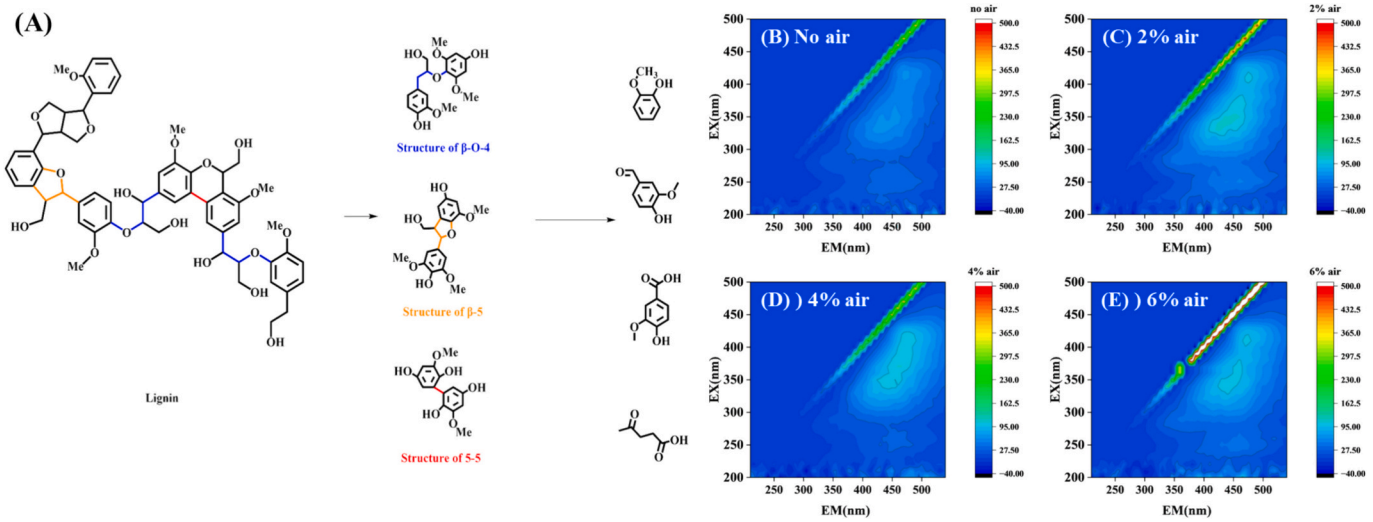


Fig. 5. (A) Depolymerization of lignin. (B–E) EEM mapping of lyophilized digestate.

bands associated with lignocellulose decomposition was significantly augmented, including O—H stretching (3400–3420 cm^{-1}), C—H vibrations of methyl and methoxy groups (2980–2840 cm^{-1}), aromatic ring skeletal vibrations (1690–1500 cm^{-1}), and C—O stretching vibrations of aromatic ethers (1050–1000 cm^{-1}). Additionally, the absorption band centered at approximately 1650 cm^{-1} was significantly intensified in the aeration groups. The band represents a composite signal attributable to aromatic C═C skeletal vibrations, asymmetric C═O stretching of carboxylate groups (COO^{-}), and C═O stretching in quinone structures [45]. The increased intensity provided strong evidence of oxidative transformation catalyzed by enzymes like laccase, leading to an increase in the aromaticity and the relative content of oxygen-containing functional groups, particularly quinone-like structures.

Further evidence was provided by EEM fluorescence analysis (Fig. 5B–E). Compared to the control group, the samples from the aeration groups exhibited higher fluorescence intensity in the characteristic regions for humic- and fulvic-like substances (Excitation/Emission: 250–400 nm/380–500 nm) [46]. The fluorescence enhancement indicated a significant increase in the content or structural complexity of fluorescent dissolved organic matter components. This finding was consistent with the FTIR observations of lignocellulose decomposition and aromatic restructuring, demonstrating that micro-aerobic conditions promoted the degree of humification.

Collectively, these findings indicated that the aeration fostered the enzymatic and radical-driven humification processes, significantly enhancing the conversion of straw lignin into humic substances during anaerobic digestion. The enhanced humification process promoted organic waste stabilization and concurrently stimulated microbial metabolic activity, thereby improving the efficiency of the digestion system.

3.5. Electrochemical analysis

The enhanced humification carried a significance in anaerobic systems that was often overlooked. To further explore potential mechanisms that promoted methane production, electrochemical properties of sludge and the role of humic substances were assessed. Cyclic voltammetry (CV) analysis (Fig. 6A) revealed significantly increased sludge capacitance under micro-aerobic conditions (2 % air: 1.187×10^{-3} F; 4 % air: 1.280×10^{-3} F; 6 % air: 1.203×10^{-3} F), representing improvements of 8.01 %, 16.47 %, and 9.46 % relative to the control group (1.099×10^{-3} F). The augmented capacitance indicated the enhancement of charge storage capability and the improvement of conductivity within the sludge matrix, which was beneficial for facilitating microbial electron transfer [47]. Correspondingly, electrochemical impedance spectroscopy (EIS) showed that the resistance of digestate samples in aeration groups (Fig. 6B) was 18.42 %–29.23 % lower than the control group, indicating the lower resistance of the sludge after the micro-aeration [48].

Humic substances, rich in quinone functional groups, are known to play a critical role as electron shuttles within microbial ecosystems, likely accelerating interspecies electron transfer and thereby enhancing anaerobic metabolisms. In this study, micro-aeration stimulated the formation of humic-like compounds within the extracellular polymeric

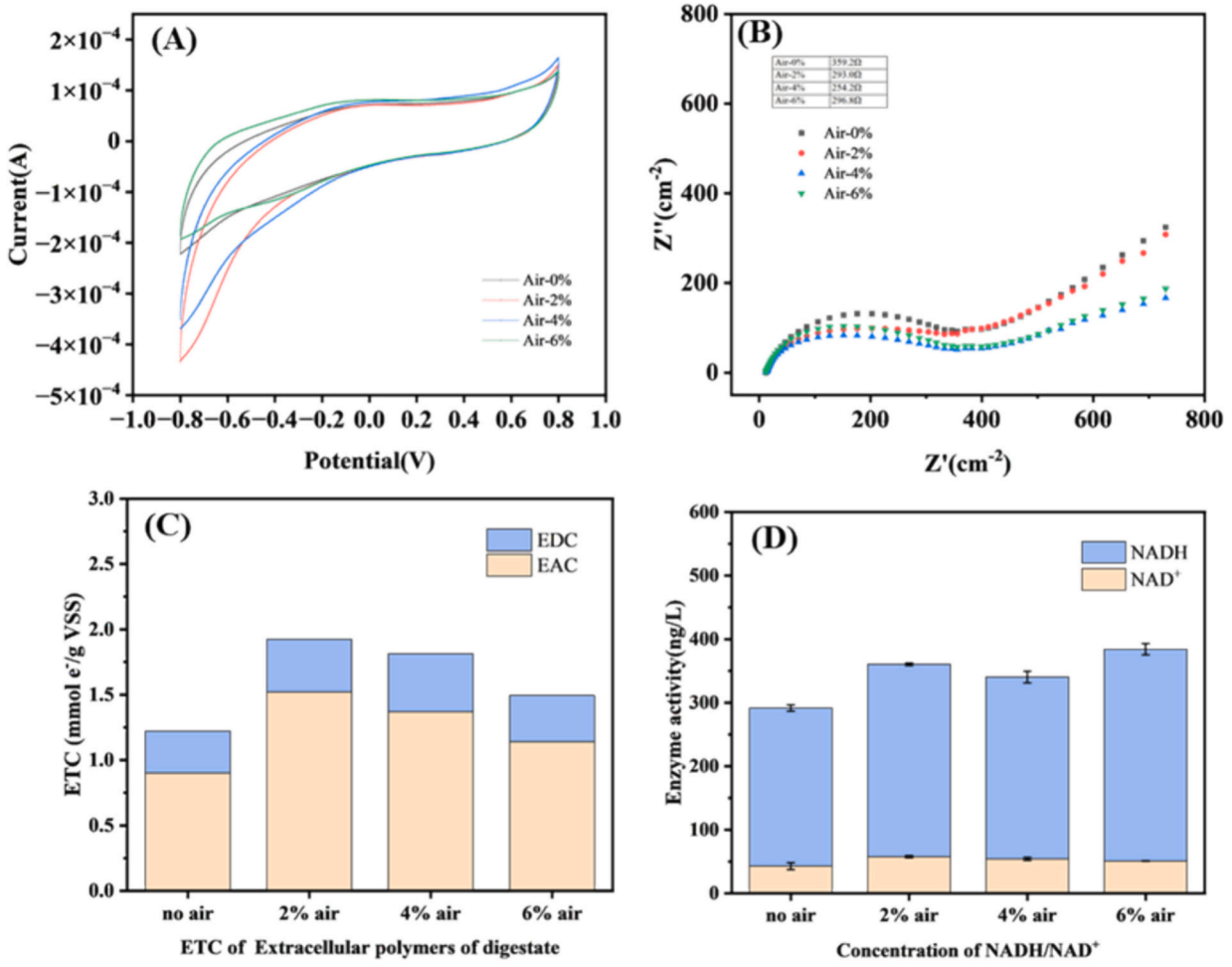


Fig. 6. Microbial electroactivity analysis: (A) CV curve. (B) Electrochemical impedance spectroscopy curves. (C) ETC of extracellular polymers. (D) Sludge intracellular $NADH/NAD^{+}$ concentration.

substances (EPS), altering their electrochemical properties.

Electron donating capacity (EAC) and electron donating capacity (EDC) were measured in EPS extracted from the digested sludge (Fig. 6C). Compared to the control group (EAC: 0.90 mmol e^-/g; EDC: 0.322 mmol e^-/g), the EAC values increased by 68.9 %, 52.2 %, and 26.7 %, respectively, and the EDC values increased by 22.1 %, 37.5 %, and 10.2 % for the 2 %, 4 %, and 6 % air groups, respectively. The elevated EAC indicated the higher humification level [49], while the increased EDC was closely associated with the higher phenolic content [50]. Thus, the rise in both EAC and EDC in the aerated groups provided additional evidence of enhanced lignocellulose decomposition, generating a higher abundance of phenolic and aromatic intermediates. The humification process under micro-aerobic conditions was likely similar to that of aerobic composting [51], where aromatic monomers were polymerized into complex humic structures with oxidation processes. While the enhanced electrochemical properties and elevated EAC/EDC values strongly suggested humic substances acted as electron shuttles to facilitate methanogenesis, it was important to acknowledge their dual role. Humic substances (particularly quinone moieties) could also serve as terminal electron acceptors, potentially competing with methanogens for electrons. In this system, the 22.57 % increase in methane production indicated that the positive effect of humic substances as electron shuttles significantly outweighed any potential negative competition, ultimately stimulating methanogenesis. Accordingly, three-dimensional (3D) fluorescence spectroscopy (Fig. 5) revealed a significantly higher characteristic fluorescence intensity in aerated samples than in the control group. Moreover, the elevated intracellular NADH/NAD^+ ratio (0.19 vs. 0.16; Fig. 6D) indicated the enhancement of redox status and electron flux, thereby promoting microbial metabolic activity [22]. Overall, electrochemical, biochemical, and spectroscopic analysis demonstrated that micro-aeration facilitated humification and microbial electron transfer, leading to the higher methane production in anaerobic digestion systems.

3.6. Validation of the fungal role and microbial dynamics

Microbial community analysis revealed that the combination of fungal augmentation and micro-aeration was critical for enhancing lignocellulose decomposition, which subsequently triggered a profound restructuring of the native anaerobic food chain (Fig. 7). Under micro-aerobic conditions, the introduced fungal strains proliferated successfully. *Phanerochaete chrysosporium* increased from a relative abundance of 4.40 % in the control group to 4.57–6.76 % in the aeration groups, while *Aspergillus niger* increased from 9.45 % to 9.91–28.64 %. Extracellular peroxidases secreted by these fungi are known to catalyze the oxidative cleavage of recalcitrant lignocellulose [52], thereby increasing substrate bioavailability for the downstream anaerobic consortia.

A primary question regarding the fungal bio-augmentation was how the inoculated strains could drive systemic enhancements given their relatively low abundance (Fig. 7C). It was hypothesized that they act as specialized catalysts. To provide a direct causal link, the results from the supplementary batch experiment were analyzed (Fig. S5). The results provided unequivocal proof of the fungal synergistic role. Specifically, the Control-MA group (4 % air, no fungi) produced low levels of ligninolytic enzymes (Laccase at 0.25 U/L). In sharp contrast, the Fungi-MA group (4 % air + fungi) produced higher levels (Laccase at 0.51 U/L). Similarly, a stronger •OH radical signal was detected in the Fungi-MA group, while the Control-MA group showed negligible ROS generation.

It confirmed that the observed enhancement was not an artifact of micro-aeration itself, but a unique metabolic consequence of the inoculated fungal consortium being activated by micro-aeration. This validated their role as highly active, specialized "catalysts", whose functional impact far outweighs their relative abundance.

Notably, the impact on the indigenous microbial community was multifaceted, reflecting a complex interplay of competitive and compensatory metabolic pathways. Key taxa in conventional methanogenic routes were adversely affected: *Methanothrix*, the dominant acetoclastic methanogen, exhibited a reduction from 54.47 % in the control group to 45.39 % under 4 % aeration. Similarly, the fermentative genus *DMER64*, related to interspecies hydrogen transfer (IHT) [53], exhibited a dose-dependent decrease with the increase of aeration intensity, suggesting that micro-aeration was unbeneficial for acetoclastic and hydrogenotrophic methanogenesis.

Conversely, this inhibitory effect was offset by the enrichment of taxa involved in rate-limiting hydrolysis and alternative methanogenic pathway. The archaeal genus *Norank_Bathyarchaeia*, known to mediate hemicellulose and cellulose decomposition, increased from 15.49 % to 23.86 % under aeration, which could accelerate the breakdown of lignocellulosic substrates [54]. Additionally, *Penicillium* is a typical aerobic lignocellulolytic fungus, exhibited an increase from 1.40 % in the control group to1.51 % under 4 % aeration [55]. Furthermore, the enrichment of Wallemia, a genus possessing antioxidant mechanisms [56], likely enhanced the system's overall tolerance to oxygen.

These shifts suggested that oxygen may modulate the metabolism of the composite microbial consortium in anaerobic environments: while temporarily inhibiting canonical methanogenic pathways, it promoted lignocellulose degradation through the enrichment of hydrolytic taxa, potentially creating a metabolic balance that enhance overall methane production.

4. Implications

This study demonstrated that integrating *Phanerochaete chrysosporium* and *Aspergillus niger* with controlled micro-aeration (4 % V_{air}/$V_{reactor}$/day) created a synergistic system for agricultural waste valorization. This strategy created transient oxidative niches sufficient to activate the fungal ROS/enzyme pathway to break the lignin barrier without causing systemic failure of the methanogenic community. The more significant mechanistic breakthrough lied in the downstream effect. In this paradigm, waste lignin is valorized into a natural mediator that facilitated electron transfer, creating a powerful synergy.

From the perspective of environmental engineering, this finding had profound implications. It presented a practical and effective strategy to enhance the efficiency and throughput of existing anaerobic digestion facilities that handle agricultural waste. By unlocking the energy potential of previously recalcitrant biomass, this method promoted the valorization of agricultural residues. Specifically, it increased the yields of renewable energy in the form of biogas and contributed to a more robust circular bioeconomy. For instance, in agricultural regions with abundant waste such as crop straw, applying this method could effectively convert the waste into valuable biogas energy, reducing the environmental burden of waste disposal and at the same time providing a clean energy source.

5. Conclusion

This study demonstrated that a microbial consortium (*Phanerochaete chrysosporium* and *Aspergillus niger*) combined with intermittent micro-aeration (optimally 4 % V_{air}:$V_{reactor}$) enhanced methane production from straw by 22.57 %. The mechanism involved micro-aeration inducing ORP fluctuations and •OH generation. This activated the consortium to secrete key lignocellulolytic enzymes (LiP, MnP, Lac), accelerating lignin cleavage and promoting humification. The resulting humic substances acted as electron shuttles, enhancing system electroactivity and stimulating methanogenesis. Microbial analysis confirmed the survival of the inoculated fungi, which, despite low relative abundance, acted as key catalysts for lignocellulose degradation. This study provided novel insights for efficient micro-aerobic digestion engineering of agricultural wastes.

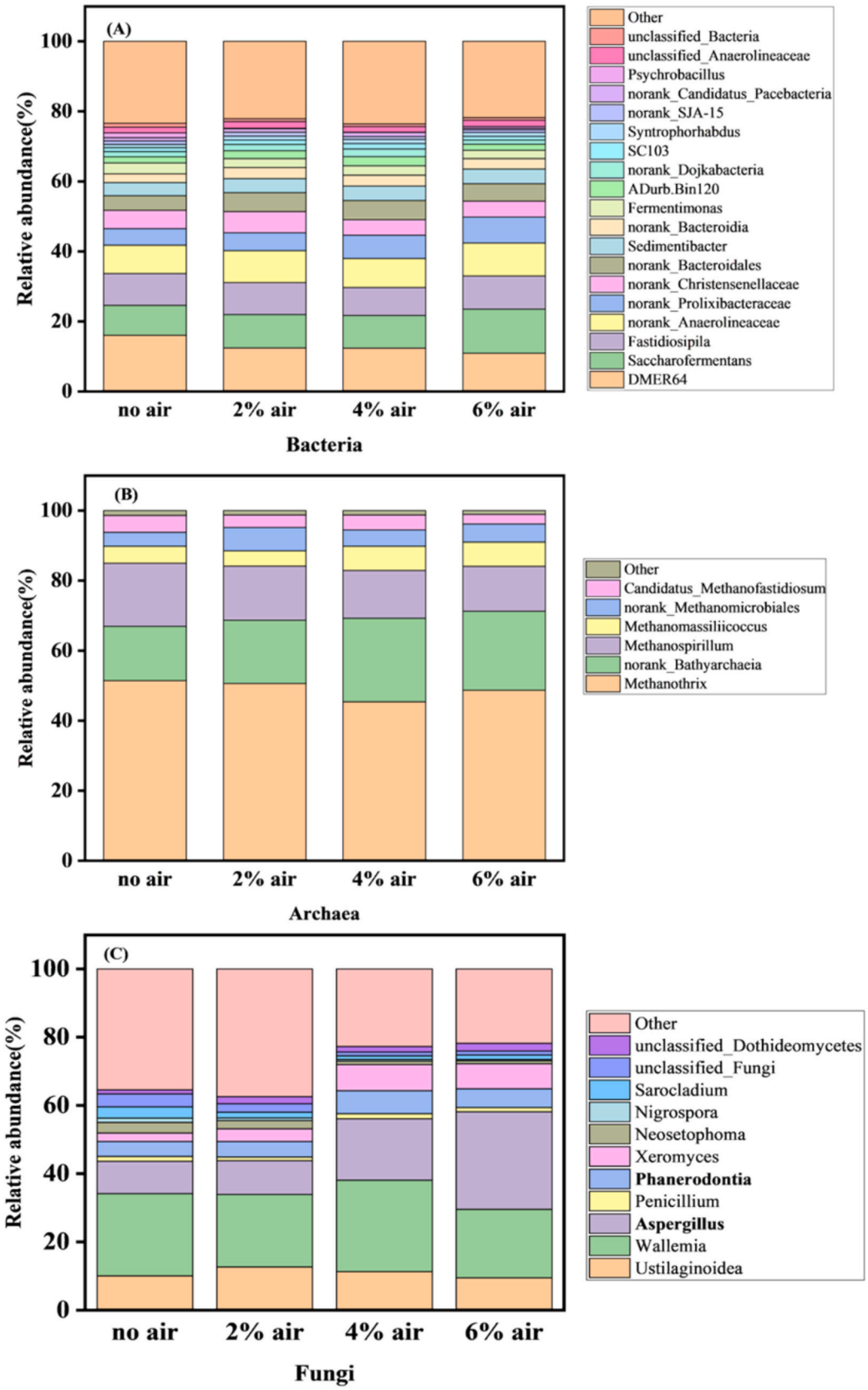


Fig. 7. Microbial communities. (A) Relative abundance of bacteria. (B) Relative abundance of archaea. (C) Relative abundance of fungi.

CRediT authorship contribution statement

Junyi Zheng: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis. **Xuepeng Wang:** Methodology, Investigation. **Yaobin Zhang:** Visualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2025.172265.

Data availability

Data will be made available on request.

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